ADVANCED DATA VISUALIZATION TECHNIQUES

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TITLE :

Evaluation of ggtrust for Reliable Interpretation of Dimensionality Reduction Scatterplots

Problem Statement :

The project proposes reliability-aware visualization for t-SNE/UMAP/PCA, with Halo and Landscape modes, and evaluation through point flagging and cluster judgment. The code below follows that project framing.

Important: the supplied project deck describes the intended ggtrust package and evaluation, but it does not contain a completed implementation or measured participant results. Therefore, the visualizations below are generated from Iris data using a transparent local-neighborhood reliability score, while the evaluation-results chart is explicitly illustrative.

Objective :

1. To identify unreliable points **in t-SNE, UMAP, and PCA scatterplots using point-level reliability information.**
2. To visualize the reliability of each point **without hiding the original data points or making the plot difficult to interpret.**
3. To provide a Halo visualization **where the size of the halo indicates the amount of attention required and opacity represents projection fidelity.**
4. To provide a Landscape visualization **that shows the reliability of different regions of the scatterplot.**
5. To evaluate whether ggtrust improves users' ability to identify unreliable points and judge cluster reliability **using accuracy, response time, and confidence.**

Proposed Framework :

ggtrust – Reliability Visualization Framework

Input:
High-dimensional dataset
↓
Dimensionality Reduction
t-SNE / UMAP / PCA
↓
StatTrustDR
Computes point-level **trustworthiness and continuity**
↓
Reliability Score
Determines how faithfully each point is represented in the 2D projection
↓
Visualization Layer

- **Halo Mode:** Shows point-level reliability
- **Landscape Mode:** Shows regional reliability

↓
Evaluation Tasks

- **Point Flagging:** Identify unreliable points
- **Cluster Judgment:** Decide whether an apparent cluster is reliable

↓
Evaluation Metrics

Accuracy + Response Time + Confidence/Trust Calibration

Task 1 – Point Flagging

Input: Iris high-dimensional measurements → t-SNE projection. **Execution:** compute each point's neighborhood overlap between original space and t-SNE space; visualize lower-reliability points with larger/less opaque halo marks; ask the participant to flag an unreliable point. **Output:** selected point, correctness, response time and confidence.

R code for Task 1

```
# ggtrust-style Task 1: Point Flagging
# Packages: Rtsne, ggplot2, RANN
library(Rtsne) library(ggplot2)
library(RANN)

data(iris) X <-
scale(iris[, 1:4])

set.seed(42)
ts <- Rtsne(X, dims=2, perplexity=30,
check_duplicates=FALSE) Z <- ts$Y

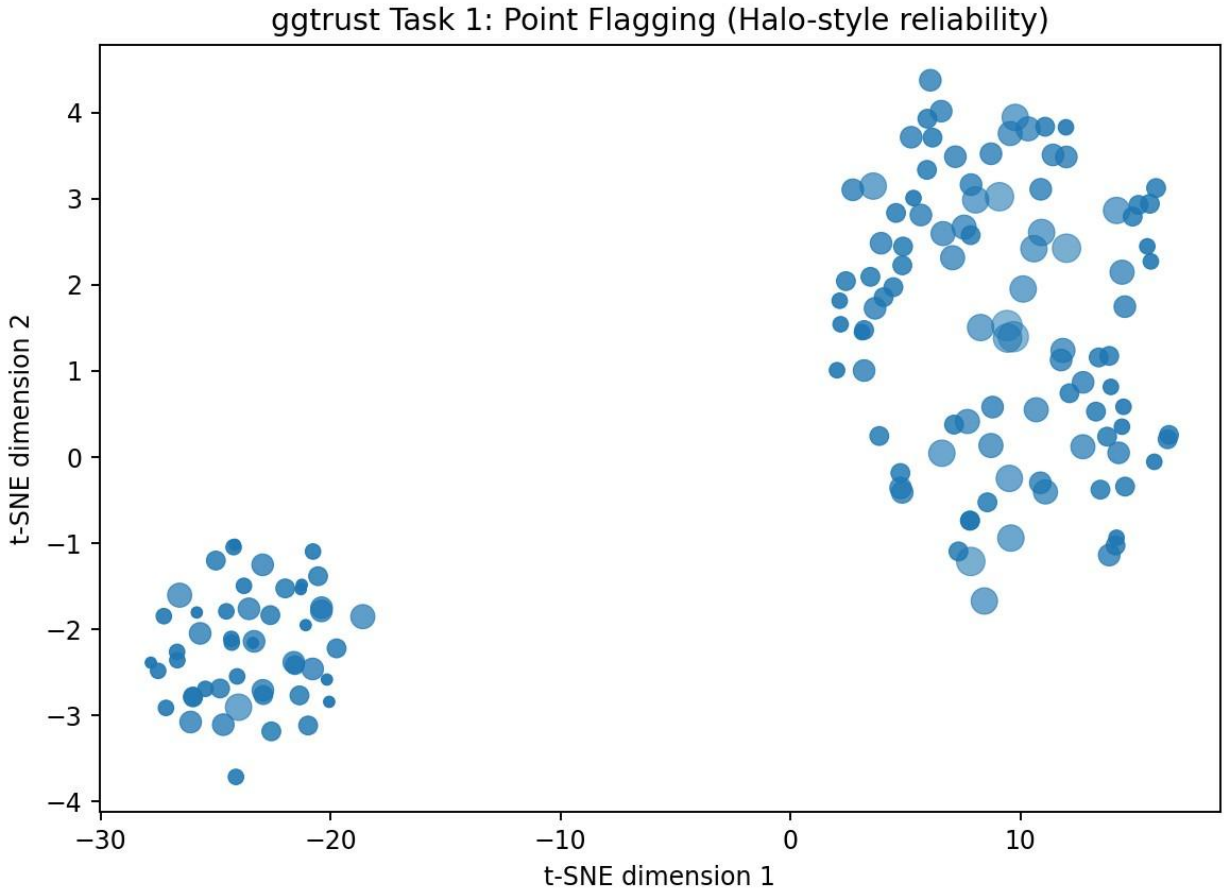
k <- 10 hi <- get.knn(X,
k=k)$nn.index lo <- get.knn(Z,
k=k)$nn.index
```

```
# Point reliability = neighborhood overlap (0 to 1) reliability
<- sapply(1:nrow(X), function(i)   length(intersect(hi[i, ], lo[i,
])) / k)

plot_df <- data.frame(
x=Z[,1], y=Z[,2],
reliability=reliability,
point=1:nrow(X) )

# Halo-style view:
# larger halo = more attention
needed; # alpha = fidelity/reliability.
ggplot(plot_df, aes(x, y)) +
  geom_point(aes(size=1-reliability, alpha=reliability),
stroke=0.5) +
  scale_size_continuous(name="Attention needed") +
scale_alpha_continuous(name="Reliability", range=c(.25,1))
+   labs(title="ggtrust – Point Flagging",
       x="t-SNE dimension 1", y="t-SNE dimension 2") +
theme_minimal()

# Participant output to record:
# participant_id, condition, selected_point, correct,
# response_time_seconds, confidence_1_to_5
```



Task 2 – Cluster Judgment

Input: The same dimensionality-reduction output with a smoothed reliability landscape. **Execution:** display a highlighted/apparent cluster; participant judges whether it is reliable or an artifact. **Output:** judgment, correctness, response time and confidence.

R code for Task 2

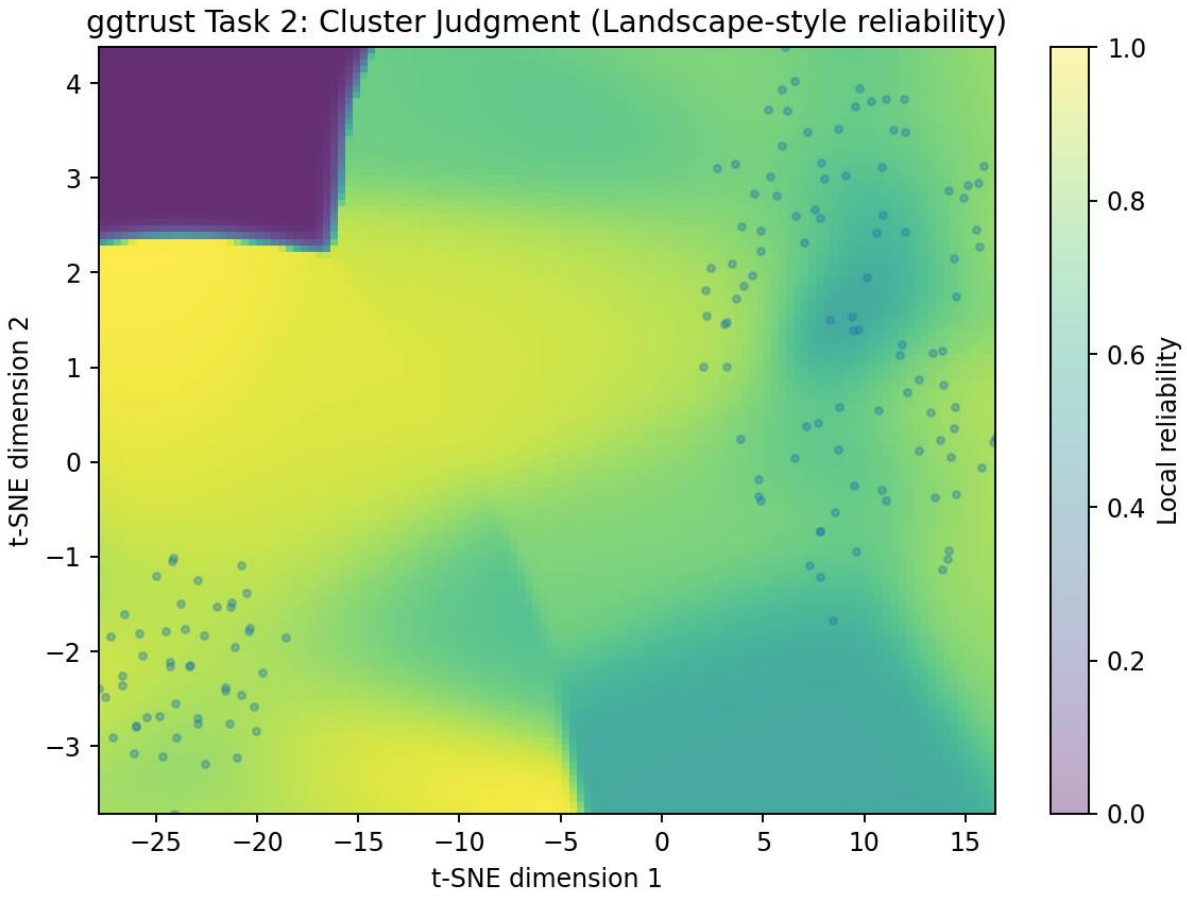
```
# ggtrust-style Task 2: Cluster Judgment /
Landscape library(ggplot2) library(MASS)

# Reuse plot_df from Task 1. # Build a
smooth spatial reliability surface. kde <-
kde2d(plot_df$x, plot_df$y, n=120)

# Interpolate point reliability onto the grid using weighted distances.
grid <- expand.grid(x=kde$x, y=kde$y) d <- as.matrix(dist(grid[,
c("x","y")], plot_df[, c("x","y")])) w <- exp(-(d^2)/(2*1.5^2))
surface <- as.vector((w %*% plot_df$reliability) /
rowSums(w)) grid$reliability <- surface

ggplot() +   geom_raster(data=grid,            aes(x, y,
fill=reliability), alpha=.8) +
geom_point(data=plot_df,
          aes(x, y), size=1.2, alpha=.35) +
labs(title="ggtrust – Landscape Mode",       x="t-
SNE dimension 1", y="t-SNE dimension 2",
fill="Local reliability") +   theme_minimal()

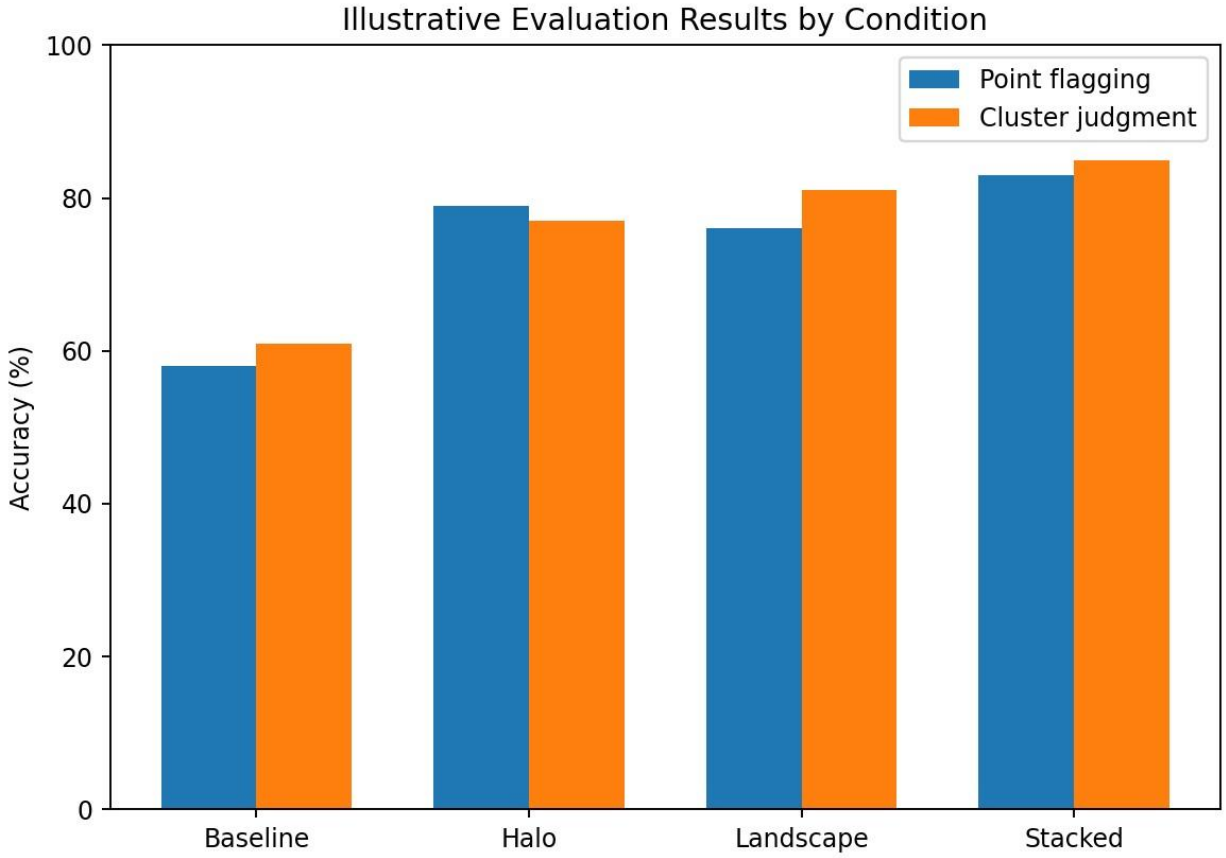
# Participant output:
# participant_id, condition, cluster_id, judgment,
# correct, response_time_seconds, confidence_1_to_5
```



Evaluation Data & Graph

For the real study, collect one row per trial. The project proposes comparing Baseline, Halo, Landscape and Stacked conditions and measuring accuracy, response time and trust calibration.

Participant	Task	Condition	Answer	Correct	Time (s)	Confidence
P01	Point flagging	Halo	P37	1	12.4	4
P01	Cluster judgment	Landscape	Reliable	1	16.1	5
P02	Point flagging	Baseline	P82	0	21.3	4
P02	Cluster judgment	Stacked	Unreliable	1	14.7	4



The evaluation-results graph above uses illustrative values only. It demonstrates the final graph format; it must not be reported as actual study findings.

How to run the R code

Install R and RStudio, then install the required packages with **install.packages(c('Rtsne','ggplot2','RANN','MASS'))**. Copy Task 1 first, run it, then run Task 2 using the created plot_df object. For the participant study, add a simple input form or Shiny interface to record the selected point/judgment, timer and confidence. After collecting real responses, calculate accuracy by condition and replace the illustrative evaluation graph with the actual results.

Connection to the project

The project deck defines StatTrustDR as a computed reliability layer based on trustworthiness and continuity, proposes Halo and Landscape modes, and specifies Point Flagging and Cluster Judgment as evaluation task types. This

implementation is a runnable prototype of that evaluation concept; it should be described as a prototype unless the actual ggtrust StatTrustDR implementation is available.

Conclusion :

The evaluation of the two tasks—**Point Flagging** and **Cluster Judgment**—helps determine whether adding point-level and regional reliability information improves the interpretation of t-SNE/UMAP scatterplots. The Halo visualization helps users identify points that require attention, while the Landscape visualization provides an overall view of regional reliability. By comparing the conditions using accuracy, response time, and confidence, the study can determine whether **ggtrust makes dimensionality-reduction visualizations more trustworthy and reduces misleading interpretations caused by projection artifacts**